



Hardware components & Time planning (for phase II)

A. HW Components for phase II:

1- Primary Processing & Microcontroller

Main Compute Unit: Raspberry Pi 4 or 5 (8GB RAM)

Role: Runs ROS 2 (Jazzy/Humble), camera vision processing (OpenCV/line detection), Lidar mapping/navigation (SLAM/Nav2), and high-level decision logic.

Storage & Cooling (ask Dr if can be eliminated) : 64GB+ High-Speed Class 10 / UHS-I MicroSD Card + Active Cooling Fan.

Low-Level Microcontroller (MCU): ESP32.

Role: Handles real-time motor PWM output, speed control, load cell ADC readings, and Emergency Stops.

2- Drive System & Motion Actuation

Drive Motors (x2): 12V DC Gear Motors with Integrated Optical Quadrature Encoders (e.g., 12V 300–500 RPM, high torque).

Motor Driver: Motor Driver Module (e.g., BTS7960 43A for high-current loads or L298N for lighter prototypes).

Front/Rear Casters (x2): Heavy-Duty Omnidirectional m Wheels (for 2WD differential drive balance).

Drive Wheels (x2):

3- Perception, Vision & Navigation Sensors

2D Lidar Scanner: RPLIDAR A1M8 or A2M8

Line Tracking & Vision Camera: Raspberry Pi Camera Module V2/V3 or USB Wide-Angle Web Camera

4- Payload & Weight Sensing System

Load Cells (x4): 5kg or 10kg Straight Bar Strain Gauge Load Cells (placed under the cargo bed corners).

Weight Amplifiers/ADCs (x2 to x4): HX711 24-Bit ADC Modules for load cell signal amplification and I2C/SPI/Serial delivery to the MCU.

Cargo Platform: Acrylic or aluminum mounting plate mounted on top of load cells to isolate total payload weight.

5. **Power Distribution & Battery Management**

Main Battery Pack: 12V or 3S LiPo Battery (e.g., 11.1V–12.8V, 5000mAh+ with high C-discharge rating).

Step-Down Voltage Regulator (Buck Converter): 12V to 5V 5A High-Current Buck Converter (e.g., XL4015 or LM2596) to power the Raspberry Pi and sensors safely without brownouts.

Power Distribution Board (PDB) / Terminal Blocks: For cleanly splitting 12V rail to motor drivers and 5V rail to Pi/Microcontroller.

- Safety Switches:

1. Master Rocker Power Switch (Toggle main power).
2. Latching Emergency Stop Push Button (E-Stop cut-off on main 12V line).
3. Inline Fuse Holder with 10A–15A Blade Fuse.

6- **Frame, Enclosure & Hardware Mechanics**

1. Main Chassis Base: Aluminum Extrusion (2020 T-Slot) or 5mm Acrylic / Laser-cut MDF dual-tier chassis.
2. Sensor & Pi Standoffs: Brass M3 M/F Standoff Kit + Nut/Bolt assortment.
3. Wiring & Interconnects: 16 AWG silicone wire (high-current motor/power lines), 24 AWG jumper wires (signals), heat-shrink tubing, zip ties, and terminal crimp connectors.

B. Timeline for phase II:

1.0 PHASE 1: HARDWARE PROCUREMENT & BENCH INTERFACING (OCTOBER)

- ├── 1.1 Procurement & Mechanical Assembly (Week 1)
 - | ├── 1.1.1 Procure 12V DC gear motors, BTS7960 drivers, RPLIDAR, HX711, battery
 - | ├── 1.1.2 Assemble 2WD differential chassis and mounting plates
 - | └── 1.1.3 Mount drive wheels, caster wheels, and strain load cells
- ├── 1.2 Power Architecture & Safety Circuitry (Week 2)
 - | ├── 1.2.1 Wire 12V primary power rail (Switch -> Fuse -> E-Stop)
 - | ├── 1.2.2 Integrate 12V-to-5V 5A step-down buck converter
 - | └── 1.2.3 Conduct voltage stability and current draw bench testing
- └── 1.3 Low-Level Microcontroller Firmware (Week 3)
 - | └── 1.3.1 Develop PWM output and directional GPIO driver code

- | |—— 1.3.2 Program hardware interrupt routines for dual quadrature encoders
- | |—— 1.3.3 Integrate 4x HX711 ADC drivers and calibrate zero/weight scales
- |—— 1.4 Compute Unit & Sensor Interfacing (Week 4)
 - |—— 1.4.1 Provision Raspberry Pi 8GB OS and ROS 2 environment
 - |—— 1.4.2 Integrate RPLIDAR node via rplidar_ros and verify RViz scan stream
 - |—— 1.4.3 Configure camera pipeline (CSI/USB) for target resolution and FPS

2.0 PHASE 2: HARDWARE-IN-THE-LOOP & SOFTWARE INTEGRATION (NOVEMBER)

- |—— 2.1 ROS 2 Serial Gateway & Motor Control (Week 5)
 - | |—— 2.1.1 Implement USB-Serial node to bridge cmd_vel to MCU
 - | |—— 2.1.2 Implement and tune low-level motor PID speed controllers on MCU
 - | |—— 2.1.3 Calculate differential drive kinematics and publish /odom and /tf
- |—— 2.2 Physical SLAM & Spatial Mapping (Week 6)
 - | |—— 2.2.1 Execute slam_toolbox / cartographer using physical Lidar data
 - | |—— 2.2.2 Map real-world test facility and build occupancy grid
 - | |—— 2.2.3 Align physical frame transforms (base_link -> laser_frame)
- |—— 2.3 Edge AI Vision & Payload Logic (Week 7)
 - | |—— 2.3.1 Package load cell data into custom /payload_weight ROS 2 topic
 - | |—— 2.3.2 Deploy ONNX/OpenCV vision classification models to Raspberry Pi
 - | |—— 2.3.3 Program weight-dependent velocity scaling logic in ROS 2 node
- |—— 2.4 Autonomous Navigation Stack Setup (Week 8)
 - |—— 2.4.1 Configure local and global costmaps in Nav2 using Lidar/Vision
 - |—— 2.4.2 Execute physical waypoint-to-waypoint autonomous navigation tests

3.0 PHASE 3: SYSTEM VALIDATION & ACADEMIC DELIVERABLES (DECEMBER)

- |—— 3.1 Field Testing & Operational Metrics (Week 9)
 - | |—— 3.1.1 Validate weight sensing accuracy under motion states

- | |—— 3.1.2 Benchmark obstacle avoidance latency and E-Stop response time
- | |—— 3.1.3 Profile continuous operating battery life under full payload
- |—— 3.2 Cable Harnessing & Enclosure Packaging (Week 10)
 - | |—— 3.2.1 Route and bundle wiring using terminal blocks and heat-shrink
 - | |—— 3.2.2 Mount acrylic/3D-printed protective covers for onboard electronics
- |—— 3.3 Final Project Documentation & Defense Prep (Week 11)
 - |—— 3.3.1 Record HD video demos of autonomous navigation & payload features
 - |—— 3.3.2 Complete Graduation Project II engineering report and schematics
 - |—— 3.3.3 Finalize slide presentation and live hardware demo setup